Business Case Study

Aerofit - Descriptive Statistics & Probability

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Colab Link:
https://drive.google.com/file/d/11QW1_XJhzg2k_j0ufWH8ZdPQNOggkbWj/view?usp=sharing

About Aerofit
Aerofit is a leading brand in the field of fitness equipment. Aerofit provides a product range including machines such as treadmills, exercise bikes, gym equipment, and fitness accessories to cater to the needs of all categories of people.

Business Problem
The market research team at AeroFit wants to identify the characteristics of the target audience for each type of treadmill offered by the company, to provide a better recommendation of the treadmills to the new customers. The team decides to investigate whether there are differences across the product with respect to customer characteristics.

1. Perform descriptive analytics **to create a customer profile** for each AeroFit treadmill product by developing appropriate tables and charts.
2. For each AeroFit treadmill product, construct **two-way contingency tables** and compute all **conditional and marginal probabilities** along with their insights/impact on the business.

Dataset
The company collected the data on individuals who purchased a treadmill from the AeroFit stores during the prior three months. The dataset has the following features:
Dataset link: **Aerofit_treadmill.csv**

Product Purchased:	KP281, KP481, or KP781
Age:	In years
Gender:	Male/Female
Education:	In years
Marital Status:	Single or partnered
Usage:	The average number of times the customer plans to use the treadmill each week.
Income:	Annual income (in $)
Fitness:	Self-rated fitness on a 1-to-5 scale, where 1 is the poor shape and 5 is the excellent shape.
Miles:	The average number of miles the customer expects to walk/run each week

Product Portfolio:
- The KP281 is an entry-level treadmill that sells for $1,500.
- The KP481 is for mid-level runners that sell for $1,750.
- The KP781 treadmill is having advanced features that sell for $2,500.

AeroFit Treadmill Case Study –Data Analysis

Import Libraries

First import the required Python libraries.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Load the Dataset

```
df = pd.read_csv("/content/sample_data/aerofit_treadmill.csv")
```

Check structure

- `<class 'pandas.core.frame.DataFrame'>`
- `RangeIndex: 180 entries, 0 to 179`
- `Data columns (total 9 columns):`
- ` #   Column         Non-Null Count  Dtype`
- `---  ------         --------------  -----`
- ` 0   Product        180 non-null    object`
- ` 1   Age            180 non-null    int64`
- ` 2   Gender         180 non-null    object`
- ` 3   Education      180 non-null    int64`
- ` 4   MaritalStatus  180 non-null    object`
- ` 5   Usage          180 non-null    int64`
- ` 6   Fitness        180 non-null    int64`
- ` 7   Income         180 non-null    int64`
- ` 8   Miles          180 non-null    int64`
- `dtypes: int64(6), object(3)`
- `memory usage: 12.8+ KB`

View first rows

```
df.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

Check statistical summary

```
df.describe()
```

	Age	Education	Usage	Fitness	Income	Miles
count	180.000000	180.000000	180.000000	180.000000	180.000000	180.000000
mean	28.788889	15.572222	3.455556	3.311111	53719.577778	103.194444
std	6.943498	1.617055	1.084797	0.958869	16506.684226	51.863605
min	18.000000	12.000000	2.000000	1.000000	29562.000000	21.000000
25%	24.000000	14.000000	3.000000	3.000000	44058.750000	66.000000
50%	26.000000	16.000000	3.000000	3.000000	50596.500000	94.000000
75%	33.000000	16.000000	4.000000	4.000000	58668.000000	114.750000
max	50.000000	21.000000	7.000000	5.000000	104581.000000	360.000000

Check missing values

```
df.isnull().sum()
```

	0
Product	0
Age	0
Gender	0
Education	0
MaritalStatus	0
Usage	0
Fitness	0
Income	0
Miles	0

dtype: int64

Insight:

- Shows datatype of columns
- Detect missing values
- Understand data ranges

Data Exploration

Number of customers per product:

```
df['Product'].value_counts()
```

	count
Product	
KP281	80
KP481	60
KP781	40

dtype: int64

Percentage distribution:

```
df['Product'].value_counts(normalize=True) * 100
```

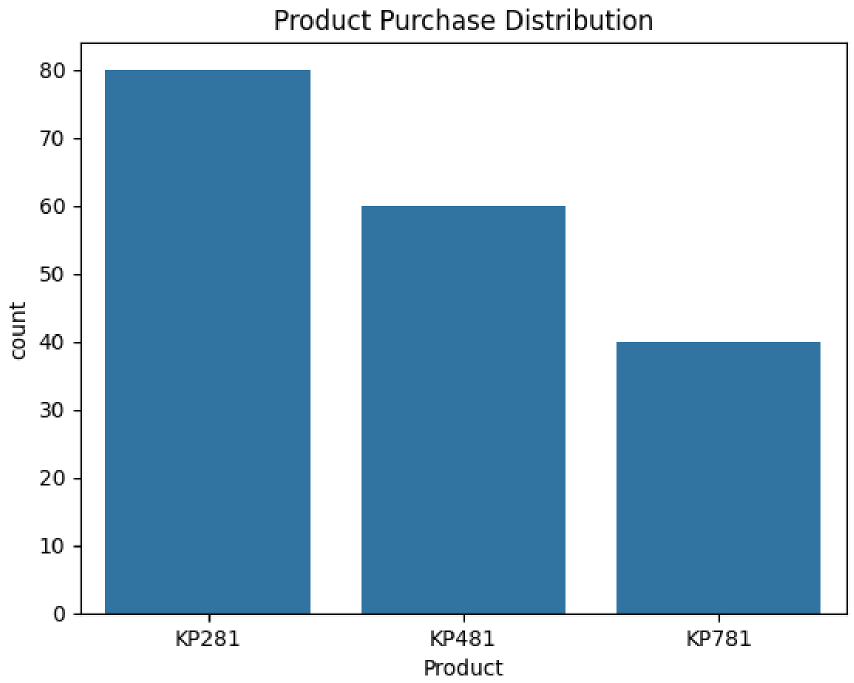
	proportion
Product	
KP281	44.444444
KP481	33.333333
KP781	22.222222

dtype: float64

Insight:
44.44 % customers buy KP281
33.33% customers buy KP481
22.22% customers buy KP781

Product Sales Visualization:

```
sns.countplot(x='Product', data=df)
plt.title("Product Purchase Distribution")
plt.show()
```

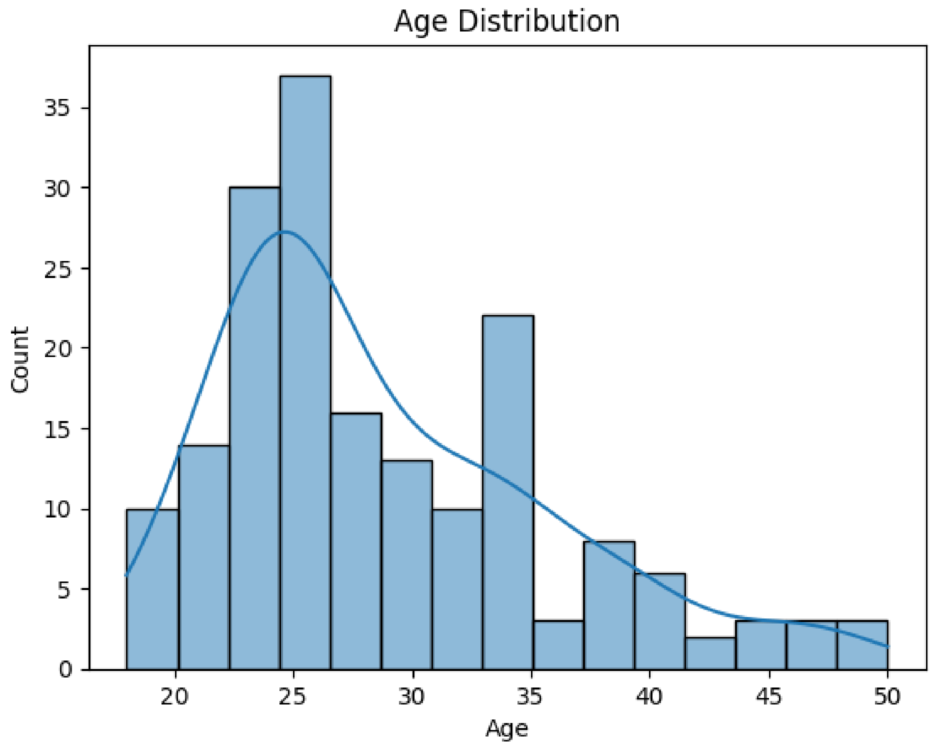


Insight:
KP281 has the highest sale.

Customer Demographics Analysis

Age Distribution

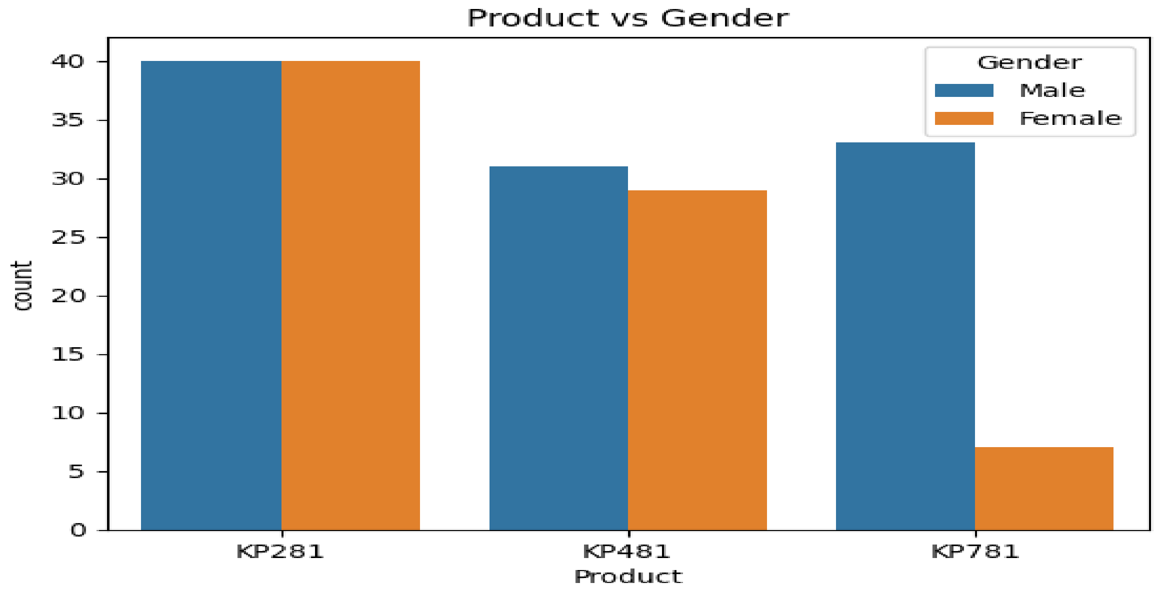
```
sns.histplot(df['Age'], bins=15, kde=True)
plt.title("Age Distribution")
plt.show()
```



Insight:
Typical age of buyers is 22 to 26.

Gender vs Product

```
sns.countplot(x='Product', hue='Gender', data=df)
plt.title("Product vs Gender")
plt.show()
```

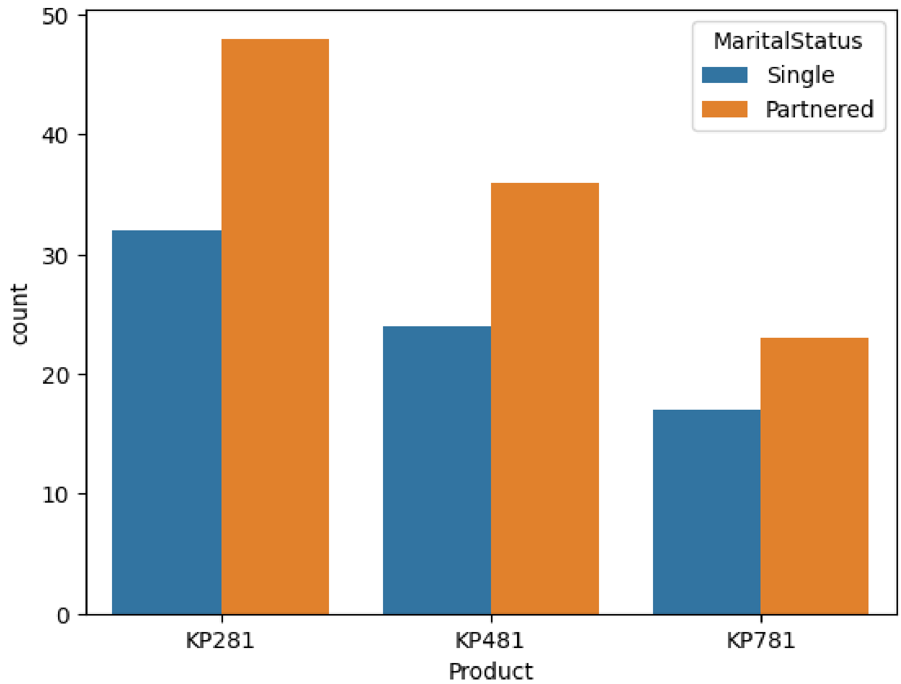


Insight:
- All products are purchased mostly by Male customers.
- KP281 has equal purchase for Male and Female.
- KP781 has the lowest sale in Female.

Marital Status vs Product

```
sns.countplot(x='Product', hue='MaritalStatus', data=df)
```

<Axes: xlabel='Product', ylabel='count'>



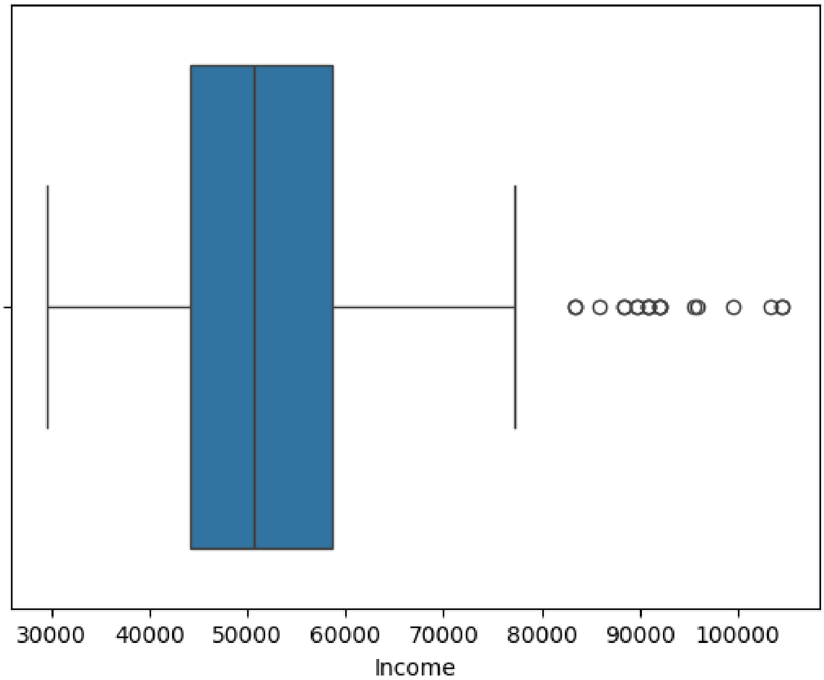
Insight:
Married people bought more products.

Outlier Detection

Boxplot:

```
sns.boxplot(x=df['Income'])
```

```
<Axes: xlabel='Income'>
```

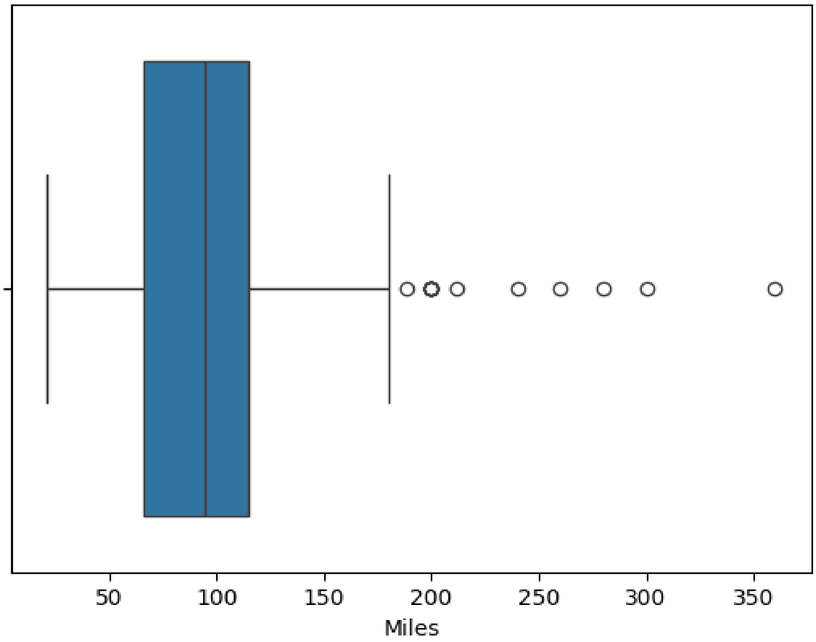


Insight:

- The income distribution shows a few high-income outliers**.**
- These customers likely belong to the premium customer segment**.**
- Such customers may prefer high-end treadmills like **KP781** with advanced features.
- These outliers indicate the presence of a luxury buyer segment in the market**.**

```
sns.boxplot(x=df['Miles'])
```

```
<Axes: xlabel='Miles'>
```



Insight:

- A few customers have significantly higher expected miles per week, indicating heavy treadmill usage.
- These users are likely fitness enthusiasts or athletes.
- These customers may prefer high-performance treadmills like KP781.

Using describe()

```
df.describe()
```

	Age	Education	Usage	Fitness	Income	Miles
count	180.000000	180.000000	180.000000	180.000000	180.000000	180.000000
mean	28.788889	15.572222	3.455556	3.311111	53719.577778	103.194444
std	6.943498	1.617055	1.084797	0.958869	16506.684226	51.863605
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75%	33.000000	16.000000	4.000000	4.000000	58668.000000	114.750000
max	50.000000	21.000000	7.000000	5.000000	104581.000000	360.000000

Insight:

Age Observation

- The average age of customers is around the early 30s.

- The minimum age is around the early 20s, and the maximum age is around the late 40s.

Insight

- Most treadmill buyers belong to the young to middle-aged adult group.
- This indicates that the primary target market for AeroFit is working professionals who are health conscious.

Education Observation

- Education ranges approximately from 12 to 21 years.
- The average education level is around 15–16 years.

Insight

- Most customers are well educated, likely college graduates.
- This suggests that the product is popular among educated professionals who value health and fitness.

Usage (Times per Week) Observation

- The average treadmill usage is around 3–4 times per week.
- Some customers plan to use it up to 7 times per week.

Insight

- Most customers intend to use the treadmill regularly as part of their weekly fitness routine.
- A few heavy users may represent serious fitness enthusiasts or athletes.

Income Observation

- Customer income ranges widely from lower income levels to high income levels.
- The mean income is higher than the median, suggesting the presence of high-income outliers.

Insight

- The dataset contains customers from different income groups.
- High-income customers may be more inclined to purchase premium treadmills like KP781.

Fitness Level Observation

- Fitness levels range from 1 to 5.
- The average fitness level is around 3.

Insight

- Most customers consider themselves moderately fit.
- This indicates that the majority of buyers are intermediate fitness users rather than beginners or professional athletes.

Miles (Expected Running Distance) Observation

- Customers expect to run around 80–100 miles per week on average.
- Some customers expect significantly higher mileage.

Insight

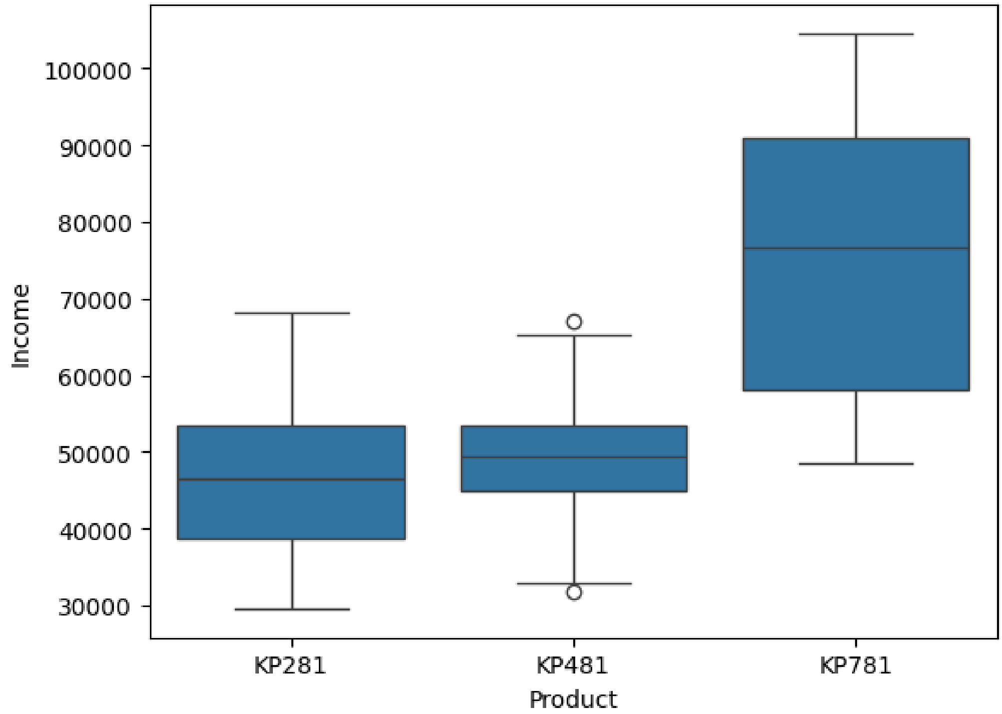
- This suggests that many users plan to use the treadmill for regular exercise and endurance training.
- High-mileage customers may prefer advanced treadmills with better durability.

The descriptive statistics show that AeroFit customers are primarily young to middle-aged, well-educated individuals with moderate fitness levels who plan to use the treadmill regularly. Income distribution indicates the presence of both mid-level and high-income customers, suggesting demand for both entry-level and premium treadmill products.

Product vs Income

```
sns.boxplot(x='Product', y='Income', data=df)
```

```
<Axes: xlabel='Product', ylabel='Income'>
```

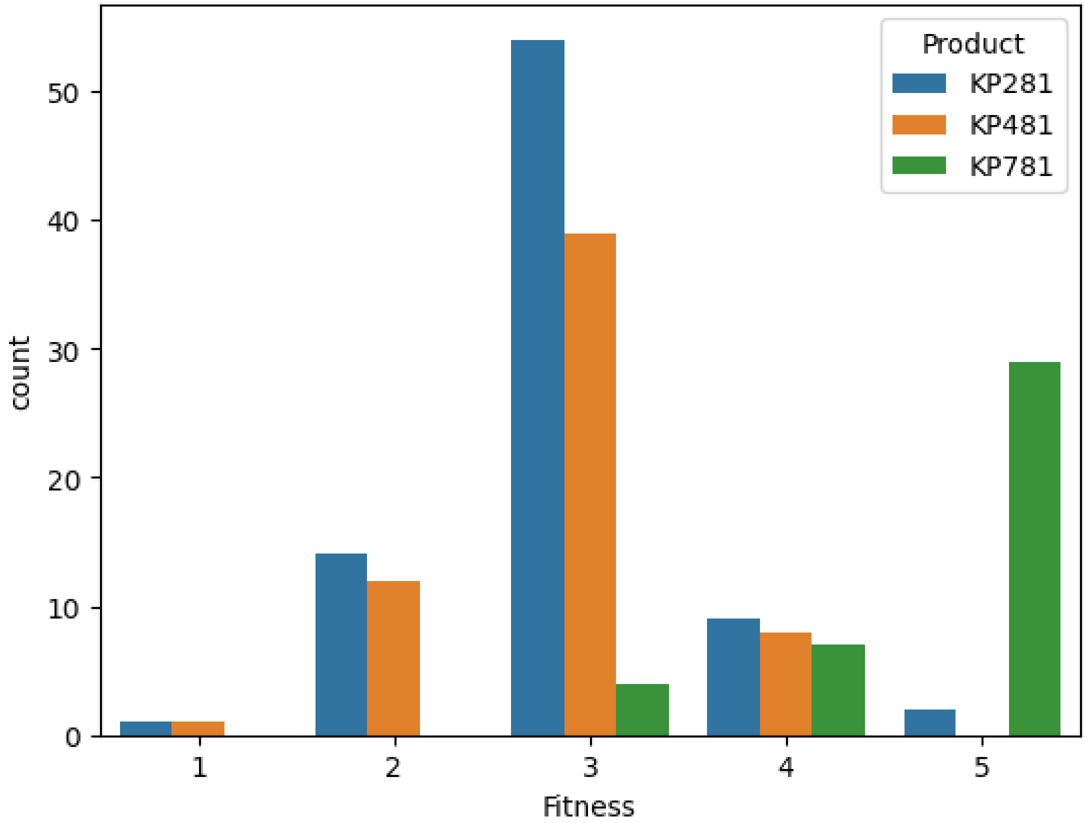


Insight:
KP781 buyers have higher income.

Product vs Fitness Level:

```
sns.countplot(x='Fitness', hue='Product', data=df)
```

```
<Axes: xlabel='Fitness', ylabel='count'>
```



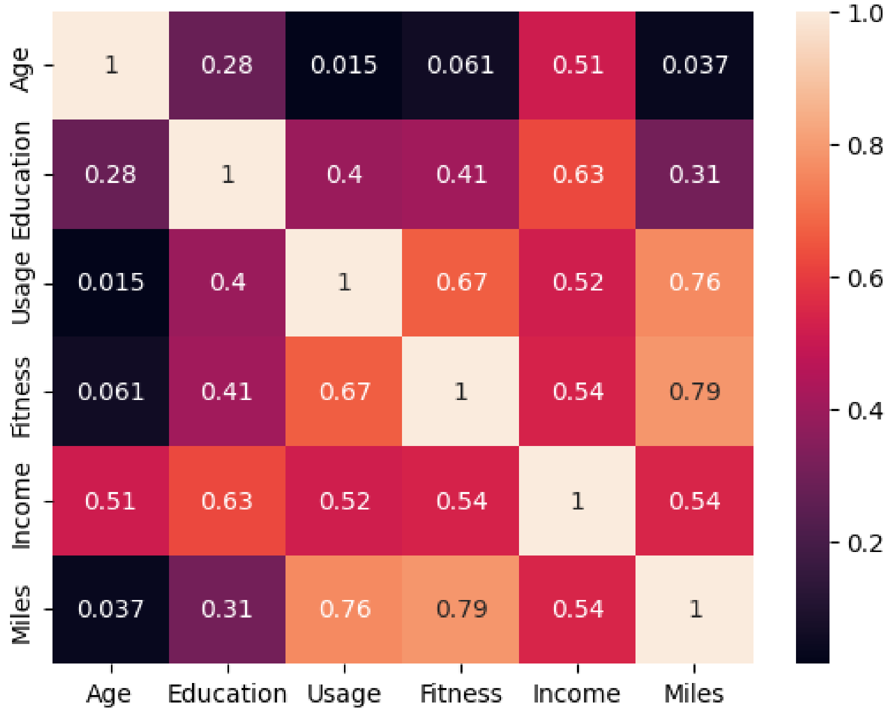
Insight:

Entry-Level Users Prefer KP281 (Fitness Level 2)
Moderately Fit Customers Prefer KP481 (Fitness Level 3)
Highly Fit Customers Prefer KP781 (Fitness Level 5)

Correlation Analysis:

```
sns.heatmap(df.corr(numeric_only=True), annot=True)
```

<Axes: >



Insight:

The correlation analysis indicates that treadmill usage and expected miles per week are strongly related, suggesting that customers who plan to exercise more frequently also expect to run longer distances. Additionally, fitness level shows a positive relationship with both usage and miles, indicating that more fit individuals tend to use the treadmill more intensively. Age and income show weaker relationships with other variables, suggesting that treadmill usage behavior is more influenced by fitness habits than demographic factors.

Product vs Gender:

```
pd.crosstab(df['Product'], df['Gender'])
```

Gender	Female	Male
Product		
KP281	40	40
KP481	29	31
KP781	7	33

Conditional Probability

Probability of Male buying KP781:

```
pd.crosstab(df['Product'], df['Gender'], normalize='columns')
```

Gender	Female	Male
Product		
KP281	0.526316	0.384615
KP481	0.381579	0.298077
KP781	0.092105	0.317308

Insight:

31% of male customers buy KP781

Product Probability given Fitness:

```
pd.crosstab(df['Fitness'], df['Product'], normalize='index')
```

Product	KP281	KP481	KP781
Fitness			
1	0.500000	0.500000	0.000000
2	0.538462	0.461538	0.000000
3	0.556701	0.402062	0.041237
4	0.375000	0.333333	0.291667
5	0.064516	0.000000	0.935484

Insight:
KP781 has highest probability for highest fitness.

Customer Profiling :

KP281 – Entry Level Buyers

Typical Profile:

- Age: 25-35
- Fitness: 2-3
- Income: Low-medium
- Usage: 2-3 times per week
- Mostly beginners

KP481 – Mid Level Buyers

Typical Profile:

- Age: 30-40
- Fitness: 3-4
- Income: Medium
- Usage: 3-4 times per week

KP781 – Advanced Buyers

Typical Profile:

- Age: 30-45
- Fitness: 4-5
- Income: High
- Usage: 4-6 times per week
- Serious runners

Overall Business Insights:

Insight 1

Most customers purchase **KP281**.

Meaning:

- Beginners dominate the market.

Insight 2

High income customers prefer **KP781**.

Meaning:

- Premium customers exist but smaller segment.

Insight 3

Fitness level strongly influences product choice.

Higher fitness → advanced treadmill i.e. **KP781**.

Insight 4

Males slightly dominate **KP781 purchases**.

Possible marketing strategy: target male athletes.

Business Recommendations

Recommendation 1

Target beginners for **KP281 marketing**

- gyms
- apartments
- home fitness beginners

Recommendation 2

Promote KP781 to high income athletes

Channels:

- running clubs
- marathon communities

Recommendation 3

Offer upgrade programs

KP281 users - KP481 after 1 year.

Recommendation 4

Personalized recommendations in offline stores

Example:

If customer profile:

- Fitness = 5
- Income = High
- Usage = 5

Recommend **- KP781**